

WEEK 1

Day 1: Introduction to Python & IDE Setup

- Installing Python & VS Code
- First Python program (`print`, `input`)
- Variables, Data Types, Operators

Day 2: Conditional Statements & Loops

- `if`, `else`, `elif`
- `for` & `while` loops
- Mini Exercises: Calculator, Even/Odd, Patterns

Day 3: Functions & Strings

- Functions with arguments & return values
- String operations & methods
- Mini Exercises: Palindrome, Reverse String

Day 4: Lists, Tuples, Dictionaries

- Basic operations, indexing, iteration
- Mini Exercises: Student Marks List, Dictionary-Based Student Info

Day 5: File Handling & Exception Handling

- Read, write, append files
- `try-except` blocks
- Debugging basics

Day 6: OOP Basics (Optional for Django Models)

- Classes, Objects, Constructors
- Mini Example: Student Class

WEEK 2

Day 7: Django Setup & Project Creation

- Install Django, create project & app
- Understand folder structure
- Run the development server

Day 8: First Django Pages & URL Routing

- Create simple HTML pages
- URL patterns & views
- Mini Exercise: Home & About Page

Day 9: Templates & Static Files

- Template inheritance (`base.html`)
- CSS inclusion
- Mini Exercise: Style Home Page

Day 10: Models & Database Setup

- Django models & migrations
- Admin panel setup
- Mini Exercise: Create `Student` Model

Day 11: Forms & Create Operation

- Django forms, `POST` requests
- Saving form data to database
- Mini Exercise: Student Registration Form

Day 12: Read Operation & Display Data

- Retrieve data from database
- Show data in table format
- Mini Exercise: Student List Page

WEEK 3

Day 13: Update Operation

- Edit form
- Save updated data
- Mini Exercise: Update Student Info

Day 14: Delete Operation

- Confirm & delete record
- Mini Exercise: Delete Student

Day 15: Search & Filter (Optional)

- Search by name or ID
- Filter records
- Mini Exercise: Search Student

Day 16: Frontend Improvements

- Use Bootstrap for styling
- Responsive table & forms
- Mini Exercise: Style Project Pages

Day 17: Debugging & Validation

- Form validations
- Debugging errors
- Mini Exercise: Validate student form

Day 18: Project Testing & Review

- Test all CRUD features
- Fix bugs
- Instructor Q&A

WEEK 4

Day 19: GitHub & Version Control

- Initialize git repository
- Push project to GitHub
- Mini Exercise: Commit & Push

Day 20: Project Deployment Basics

- Introduction to hosting (PythonAnywhere/Heroku)
- Configure project for deployment

Day 21: Final Touches

- Final UI improvements
- Add project documentation

Day 22: Project Presentation Preparation

- Prepare slides/report
- Explain project flow

Day 23: Mock Presentation & Feedback

- Student presents project

Day 24: Final Project Presentation & Review

- Project demonstration